

1 Data Generation

1.1 Pattern Matching

A *pattern* is a set.

1.1.1 Example

Let $a \circ b = [a_1, \dots, a_n, b_1, \dots, b_m]$, where $a = [a_1, \dots, a_n]$ and $b = [b_1, \dots, b_m]$.

$$\begin{aligned}\mathfrak{E} &:= \mathfrak{T} \cup \{\mathfrak{e}_1 \circ \mathfrak{o} \circ \mathfrak{e}_2 \mid \mathfrak{o} \in \{[“+”], [“-”]\} \wedge \mathfrak{e}_1, \mathfrak{e}_2 \in \mathfrak{E}\} \\ \mathfrak{T} &:= \mathfrak{F} \cup \{\mathfrak{t}_1 \circ \mathfrak{o} \circ \mathfrak{t}_2 \mid \mathfrak{o} \in \{[“\times”], [“\div”]\} \wedge \mathfrak{t}_1, \mathfrak{t}_2 \in \mathfrak{T}\} \\ \mathfrak{F} &:= \mathfrak{N} \cup \{[“(”] \circ \mathfrak{e} \circ [“)”] \mid \mathfrak{e} \in \mathfrak{E}\} \\ \mathfrak{N} &:= N \cup N^* \\ N &:= \{“0”, \dots, “9”\}\end{aligned}$$

What is the difference between pattern matching and tokenwise parsing?

2 gen.jl and pattern.jl

A *grimoire* of a pattern $\mathfrak{P}$ is a function $\Phi : L_\Phi \rightarrow \mathcal{P}(\mathfrak{P})$, where $L_\Phi \subseteq \mathcal{P}(\mathfrak{P})$ is a set of *labels*¹.

Let $G := (\langle \mathcal{G} \rangle, *)$ be a semigroup. Let $\Delta : [\Phi] \times G^2 \rightarrow [\Phi]$ – where $[\Phi]$ is the set of all grimoires of a pattern $\mathfrak{P} \subseteq G^{<\omega}$ – be defined for some grimoire Φ by $\forall l \in L_\Phi \setminus \{l_1 * l_2\} \ (\Phi'(l) = \Phi(l))$ and $\Phi'(l_1 * l_2) = \Phi(l_1 * l_2) \cup \{[l_1, l_2]\}$ if $l_1 * l_2 \in L_\Phi$ and $\Phi'(l_1 * l_2) = \{[l_1, l_2]\}$ if not – where $\Phi' = \Delta(\Phi, [l_1, l_2])$ and $l_1, l_2 \in G$. Note that since $L_\Phi = \text{dom}(\Phi)$, then $L_{\Phi'} = L_\Phi \cup \{l_1 * l_2\}$ if $\Phi' = \Delta(\Phi, [l_1, l_2])$.

Now, consider the exploration function $\mathcal{E} :$

$[\Phi] \times [\mathcal{C}]^{<\omega} \rightarrow [\Phi]$ defined by $\mathcal{E}(\Phi, [\mathcal{C}_1, \dots, \mathcal{C}_n]) := \bigcirc_{i=1}^n \Delta(\Phi, \mathcal{C}_i(L_\Phi))$

is another subgroup!!!, where $[\mathcal{C}] \subset \mathcal{P}(G) \rightarrow G^2$ and $\mathcal{C}(S) \in S^2$ for each $\mathcal{C} \in [\mathcal{C}]$ and $S \in \mathcal{P}(G)$. Then Φ_n is a partial or full representation of the group G , where each element in $L(\Phi_n)$ maps to the set of ordered pairs of elements in G to which it is mapped by $*$.

¹Labes – or, *concepts* – are useful abstractions of patterns, which will be used later to optimize the complexity of solution-finding systems. The subsets $\mathfrak{T}, \mathfrak{F}, \mathfrak{N}$ and N in example 1.1.1.1 are examples of labes.

!!!!!!! therefore, for each output generated by $\mathcal{E}$, one can define a function λ such that: !!!!!!! Let the functions $\lambda, \lambda_{\text{step}} : X_G \rightarrow \mathcal{G}^{<\omega}$ – where

$$X_G = \left\{ [\Phi, \sigma, \Pi, T] \in (L_\Phi \rightarrow \mathcal{P}(G^{<\omega})) \times G^{<\omega} \times \boxed{\Pi} \times \mathcal{P}(G) \mid \sigma \in L_\Phi^{<\omega} \wedge T \subseteq L_\Phi \wedge \forall$$

– be defined by

$$\lambda(\Phi, \sigma, \Pi, T) := \begin{cases} \sigma & : \lambda_{\text{step}}(\Phi, \sigma, \Pi, T) = \sigma \\ \lambda(\Phi, \lambda_{\text{step}}(\Phi, \sigma, \Pi, T), \Pi, T) & : \text{otherwise} \end{cases}$$

and

$$\lambda_{\text{step}}(\Phi, \sigma, \Pi, T) := \bigotimes_{i=1}^{|\sigma|} \left(\begin{cases} \sigma_i & : \sigma_i \in T \\ \Pi(\Phi(\sigma_i)) & : \text{otherwise} \end{cases} \right),$$

where $\boxed{\Pi} \subset \mathcal{P}(G^{<\omega}) \rightarrow G^{<\omega}$ is a set of maps wherein $\Pi(S) \in S$ for each $\Pi \in \boxed{\Pi}$ and $S \in \mathcal{P}(G^{<\omega})$ hold.

!!!!!!BEVISA FAKTUM OM DEN GENERERADE STRUKTUREN.

SKRIV EXEMPEL FÖR ATT DEMONSTRERA HUR EN GRIMOIRE BYGGS UPP AV $\mathcal{E}$, OCH SEDAN HUR λ GENERERAR EN LÖSNING – T.EX. FÖR EN PERMUTATIONSGRUPP (ELLER NÅGON ANNAN SYNTAX).!!!!!!